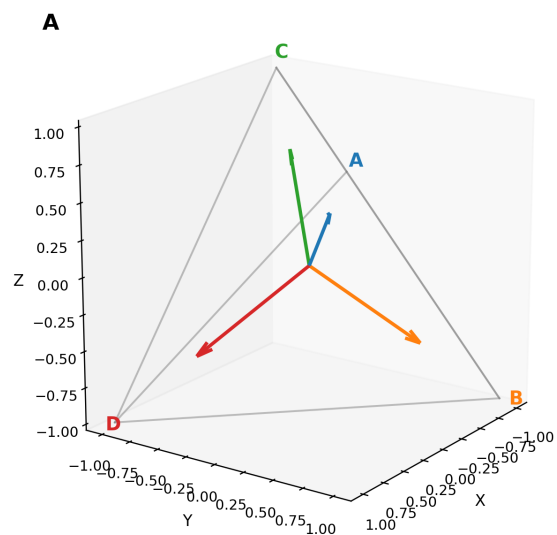
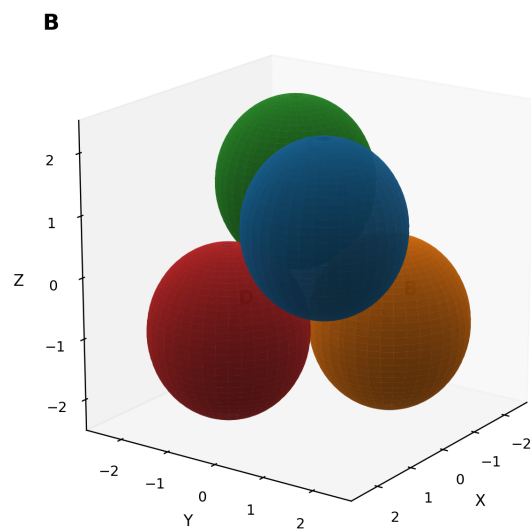




A — Quadray axes (A,B,C,D)

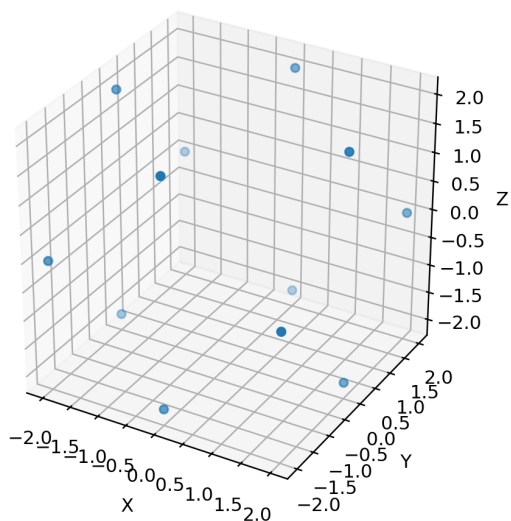


B — Tetrahedron as close-packed spheres (IVM/CCP/FCC)

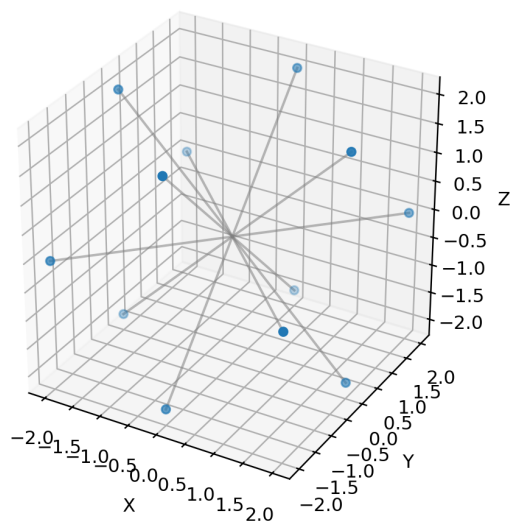




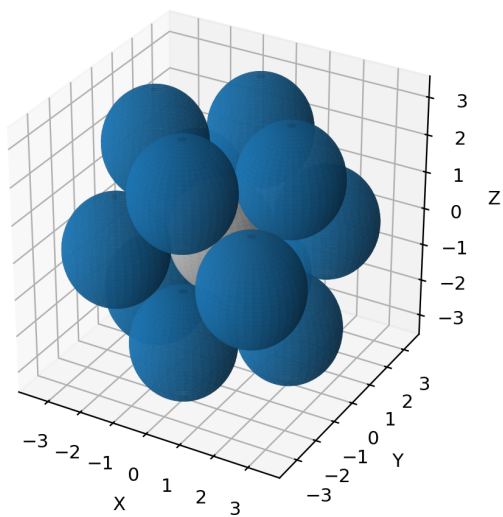
A — IVM neighbors (points)



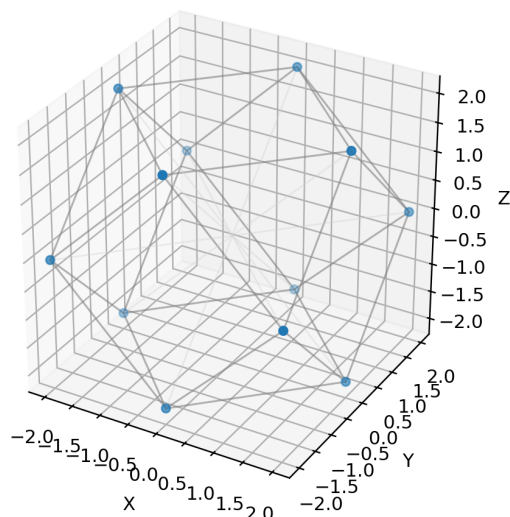
B — Neighbors with radial edges



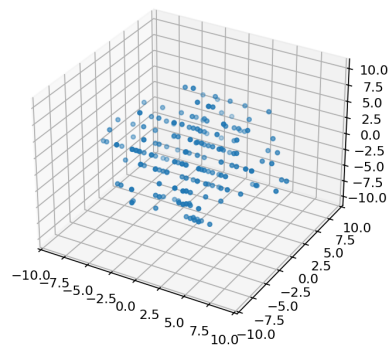
C — Twelve around one (close-packed)



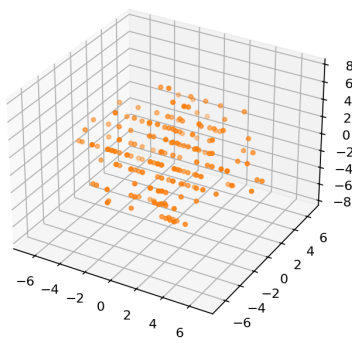
D — Vector equilibrium (adjacency)



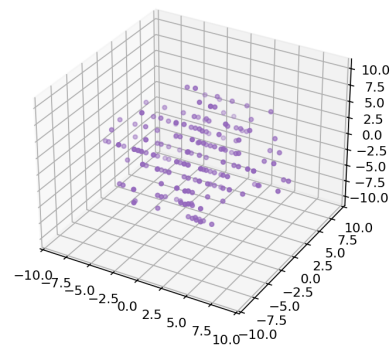
Default embedding



Scaled embedding (0.75x)



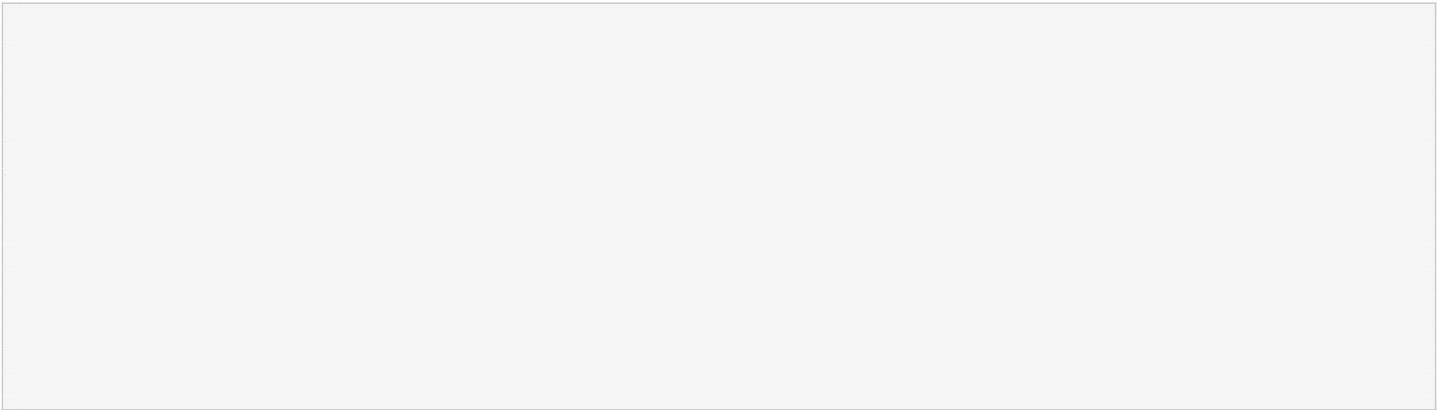
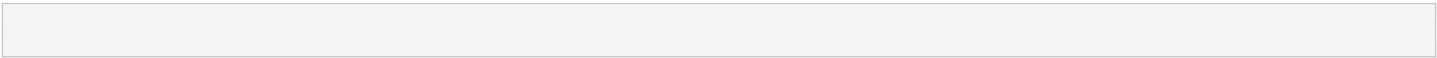
Urner embedding



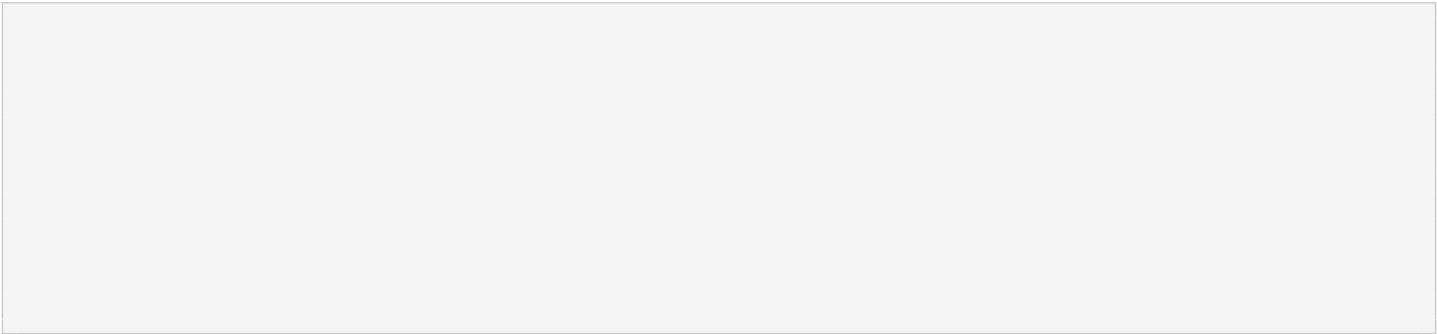
$$\frac{1}{4}$$

$$\frac{1}{4}$$

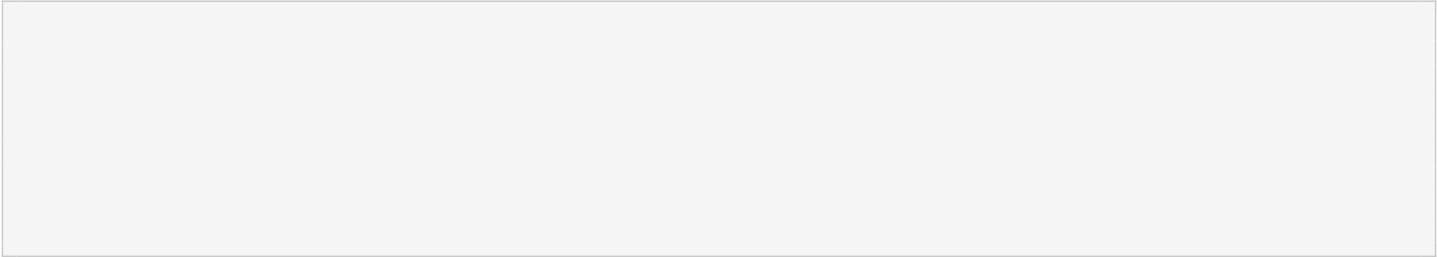
$$(2, 1, 1, 0)$$



$$\pm 2\hat{e}_i 2\sqrt{2}\frac{1}{2}$$



$$(\pm 1, \pm 1, \pm 1)\frac{1}{2}$$



$$(k, k, k, k)$$

$$\begin{array}{l}||/4\\(2,1,1,0)\end{array}$$

$$(0,0,0,0)(1,0,0,0)(0,1,0,0)(0,0,1,0)\frac{1}{4}$$

$$\frac{P_0,\dots,P_3}{VS3=\sqrt{9/8}}$$

$$\begin{array}{cc} |_{5\times 5} & |_{3\times 3} \\ & | \hspace{0.5cm} | /4 \end{array}$$

$$\frac{1}{4}$$

$$(P_1-P_0)$$

$$d_{ij}P_iP_j$$

$$V_{ivm}=S3\cdot V_{xyz}S3=\sqrt{9/8}$$

$$V_{ivm}=S3\cdot V_{xyz}S3=\sqrt{9/8}$$

$$P_1-P_0P_2-P_0P_3-P_0\frac{1}{4}$$

$$[i,j]=\mathbb{E}[\partial_{\theta_i} \; p(x;\theta) \; \partial_{\theta_j} \; p(x;\theta) \,]$$

$$_{288}V^2=\begin{pmatrix}0&1&1&1&1\\1&0&d_{01}^2&d_{02}^2&d_{03}^2\\1&d_{10}^2&0&d_{12}^2&d_{13}^2\\1&d_{20}^2&d_{21}^2&0&d_{23}^2\\1&d_{30}^2&d_{31}^2&d_{32}^2&0\end{pmatrix}$$

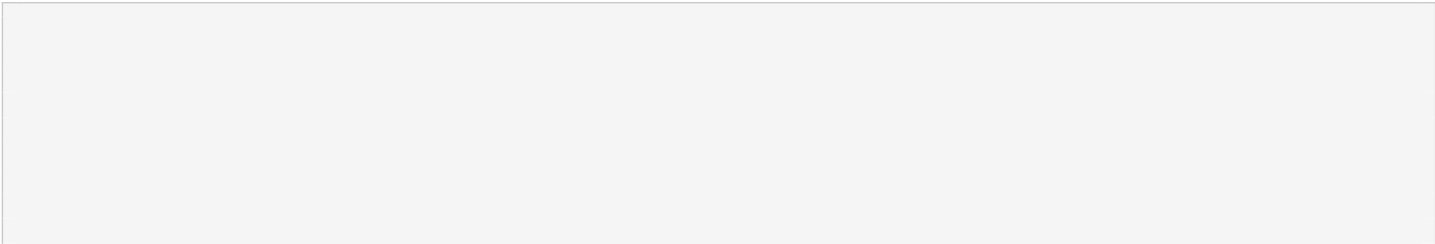
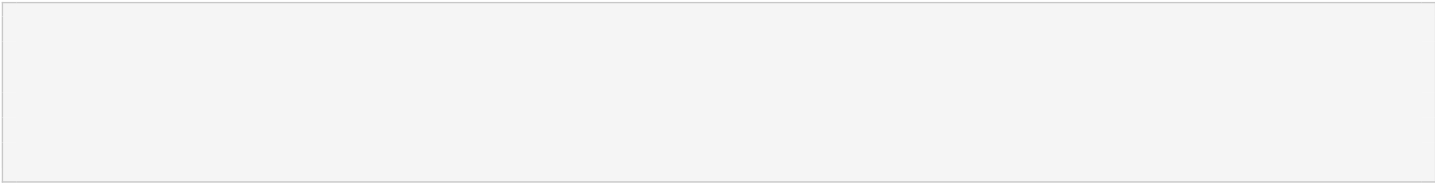
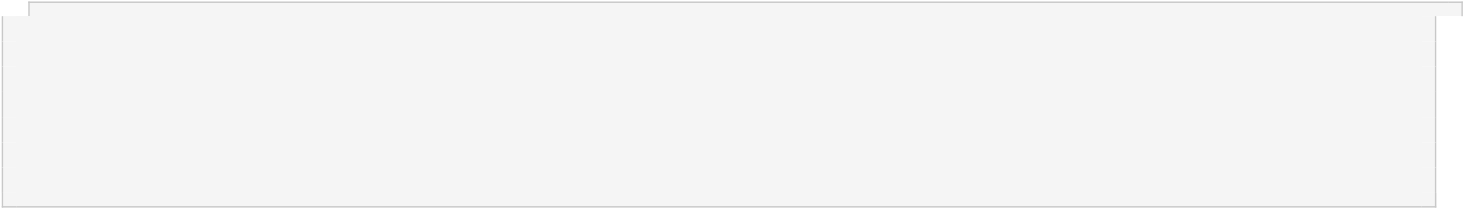
$$S3=\sqrt{9/8}$$

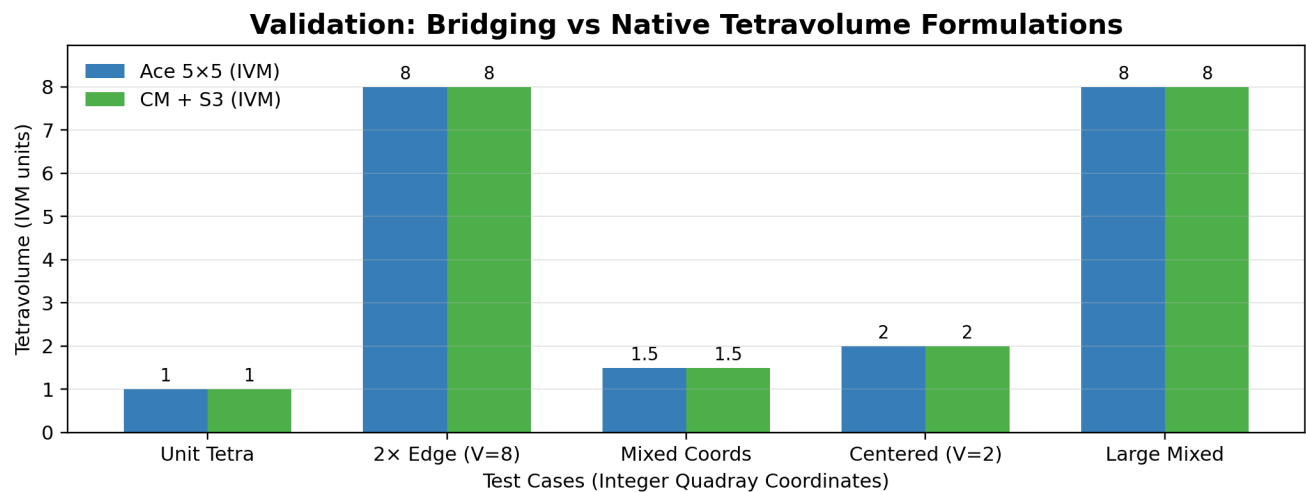
$$a,b,cd,e,f$$

$$V_{ivm}=S3\cdot V_{xyz}S3=\sqrt{9/8}$$

$$P_1-P_0P_2-P_0P_3-P_0\frac{1}{4}$$

$$S3=\sqrt{9/8}$$



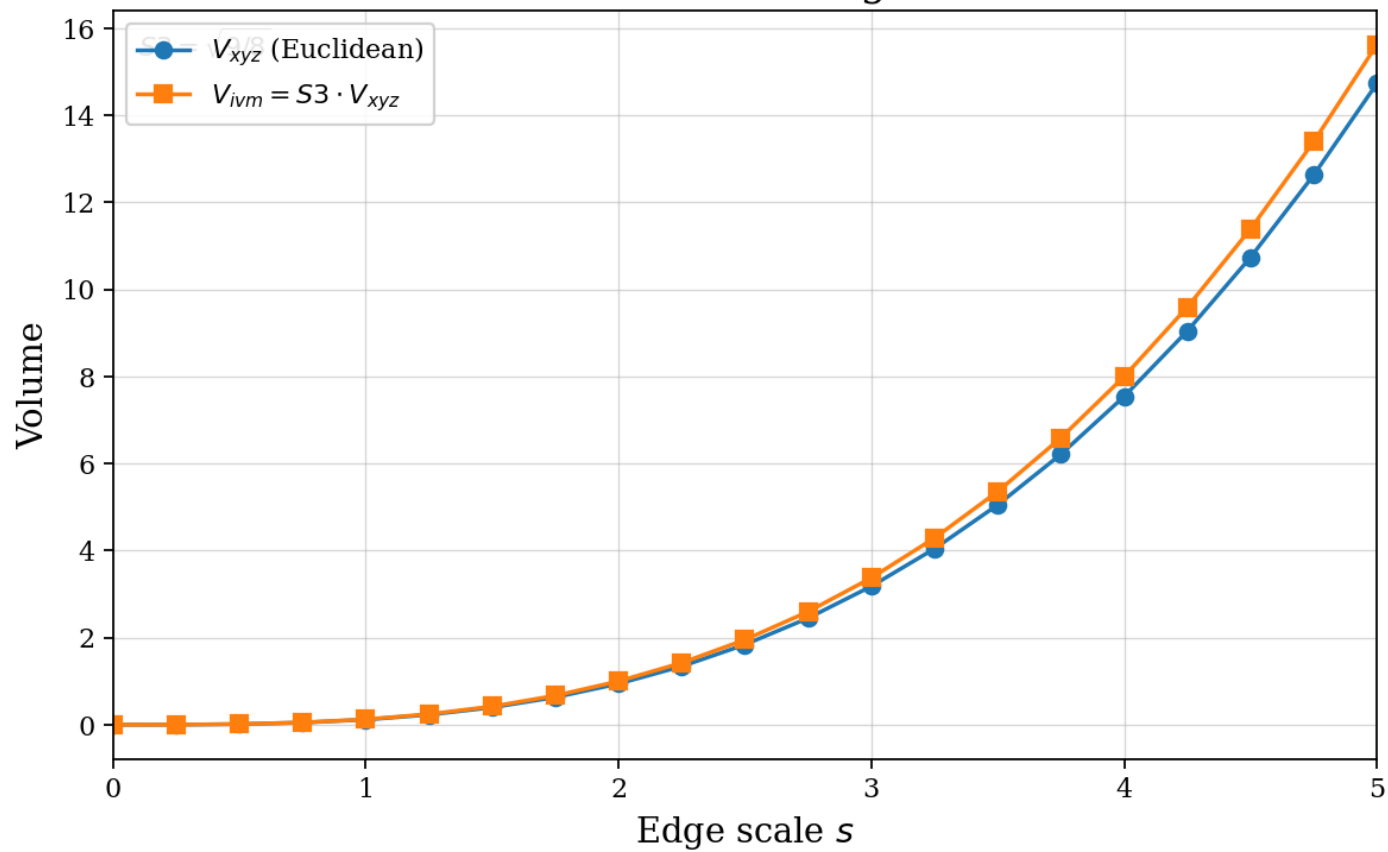


Test cases: Unit tetra (V=1), 2× edge scaling (V=8), mixed coordinates, centered (V=2), large mixed.  
 Both methods agree exactly (symbolic identity), validating the S3 conversion factor. Data: quadmath/output/data/bridging\_vs\_native.csv

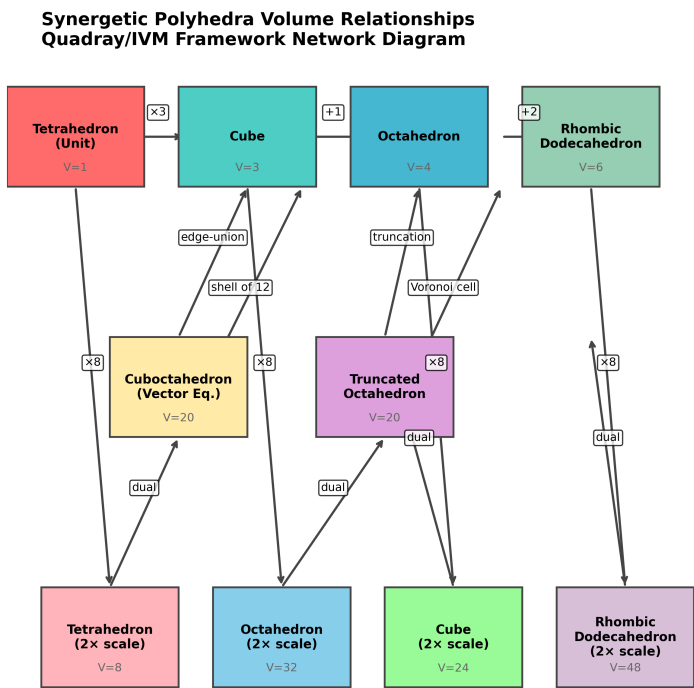
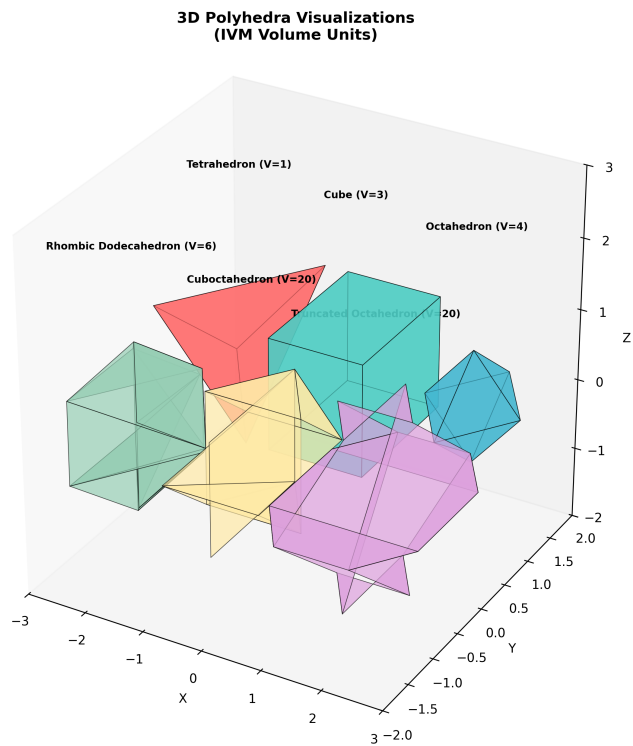
$$S3 = \sqrt{9/8}$$

(2, 1, 1, 0)

Tetra volume vs edge scale



$$V = \frac{\sqrt{2}}{12} \cdot L^3 V_{ivm} = \frac{1}{8} \cdot L^3 S3 = \sqrt{9/8} \approx 1.06066$$



Volume relationships in IVM tetra-units. Regular tetrahedron (V=1) is the fundamental unit.  
 Scaling relationships: 2x edge length → 8x volume ( $V \propto L^3$ ).  
 Geometric constructions: edge-union, truncation, dual relationships, and Voronoi cells.  
 All polyhedra constructed with consistent edge lengths and integer Quadray coordinates.



$A, B, CABC$   
 $(P_i - P_0)$

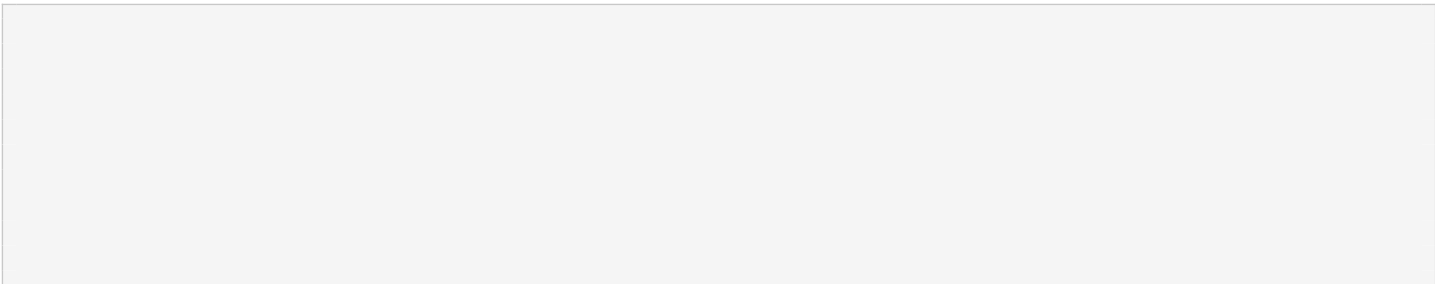
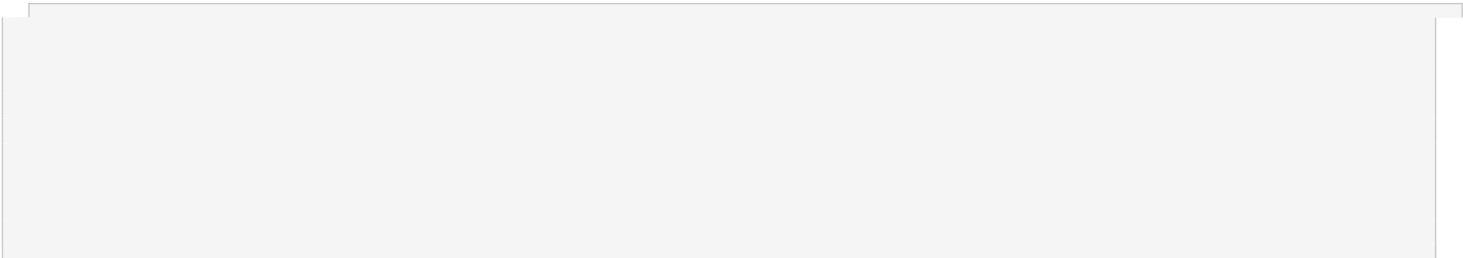


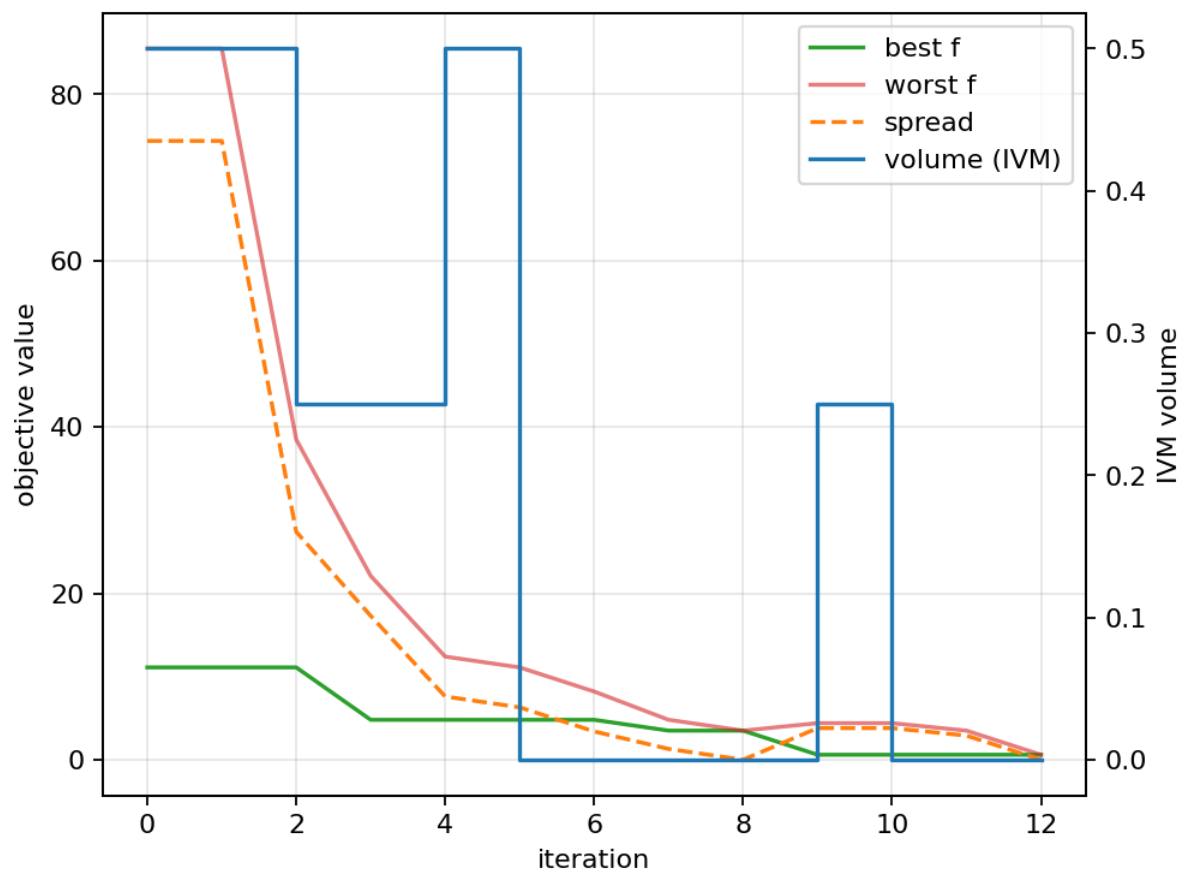
$3 \times 3 \mid /4$

*G*



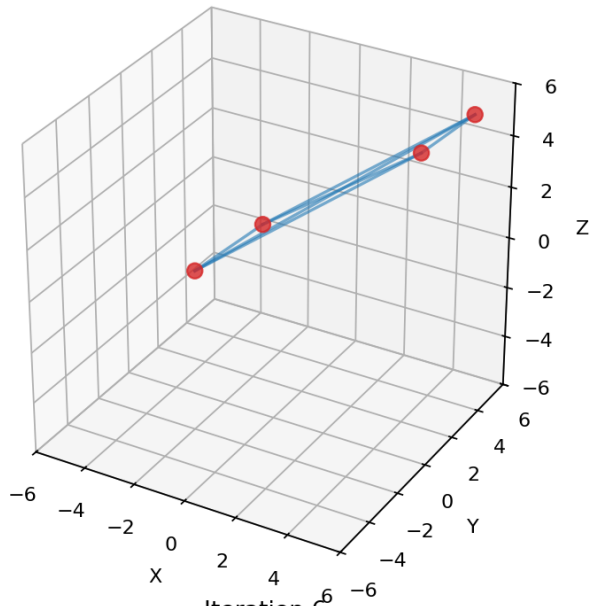
$\alpha \approx 1$   
 $\gamma \approx 2$   
 $\rho \approx 0.5$   
 $\sigma \approx 0.5$



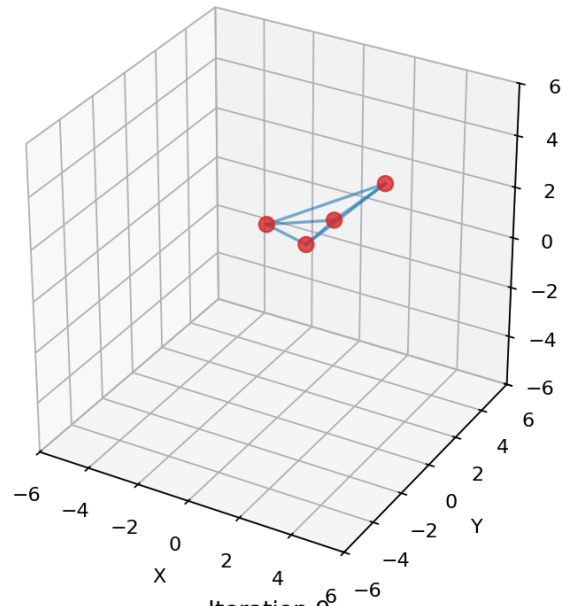


## Nelder-Mead Simplex Evolution on Integer Quadray Lattice

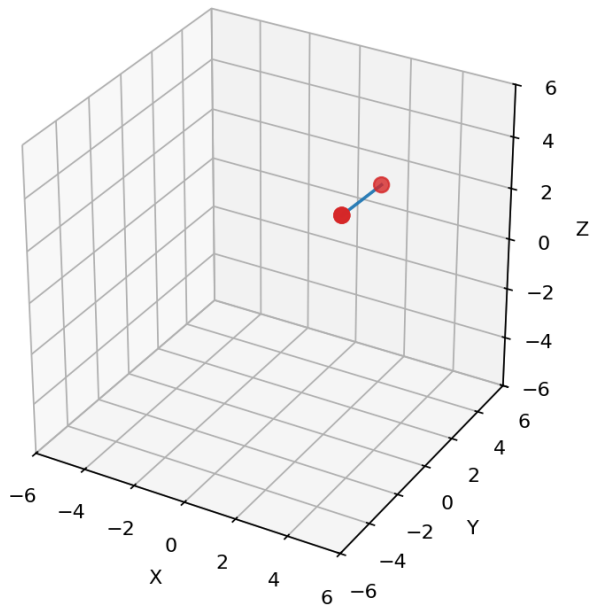
Iteration 0  
Best: 11.1, Spread: 74.4



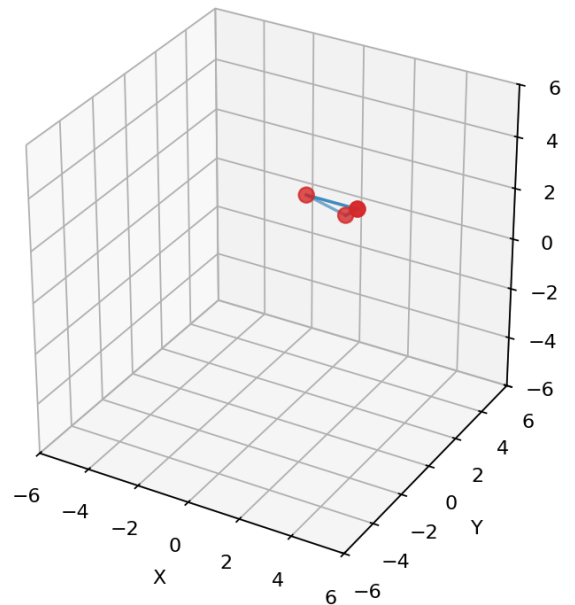
Iteration 3  
Best: 4.8, Spread: 7.6



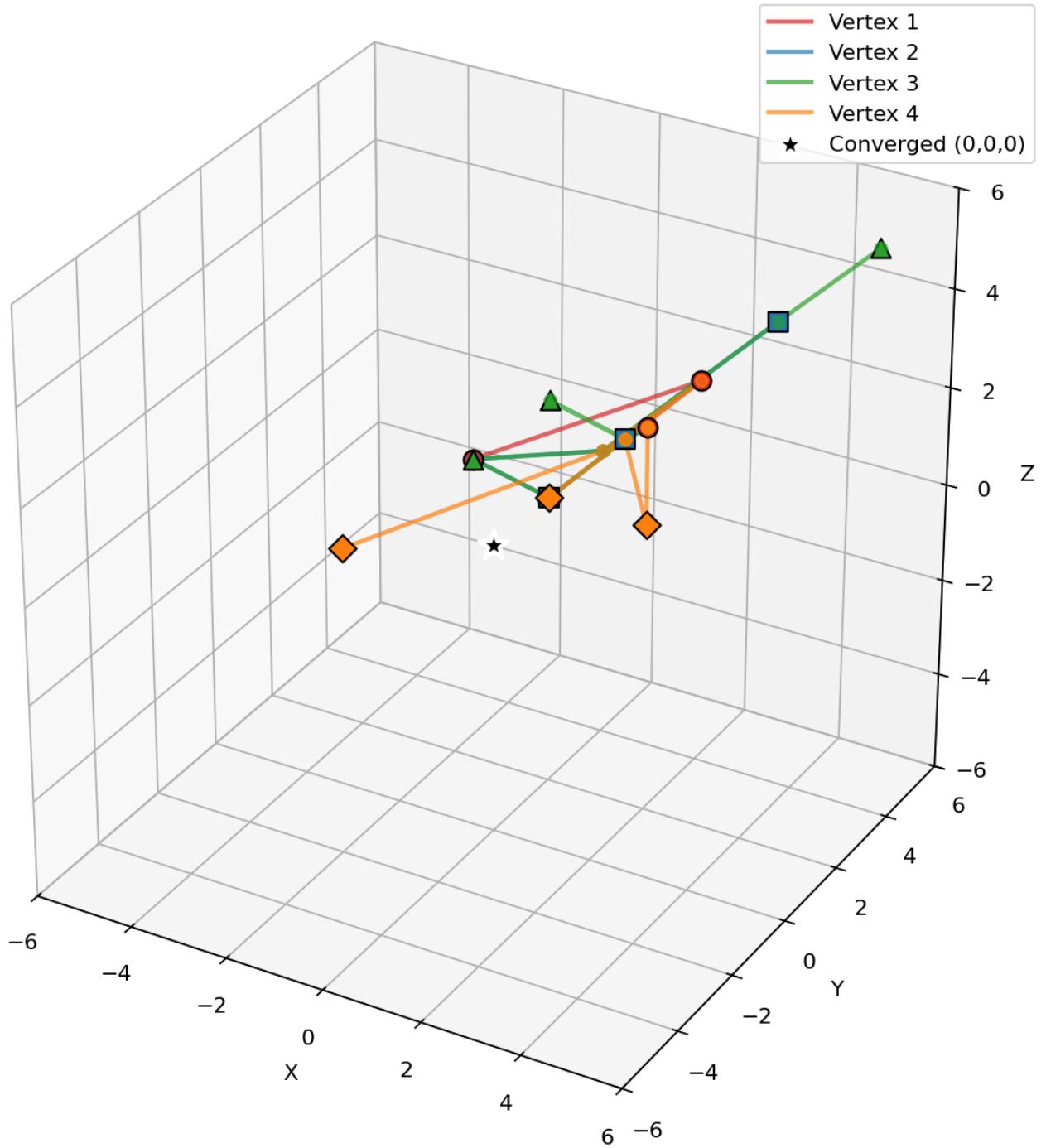
Iteration 6  
Best: 3.5, Spread: 1.3



Iteration 9  
Best: 0.6, Spread: 3.8



Complete Simplex Optimization Trace  
Nelder-Mead on Integer Quadray Lattice



$$F_{ij}$$

$$=(w_0,w_1,w_2)L()$$

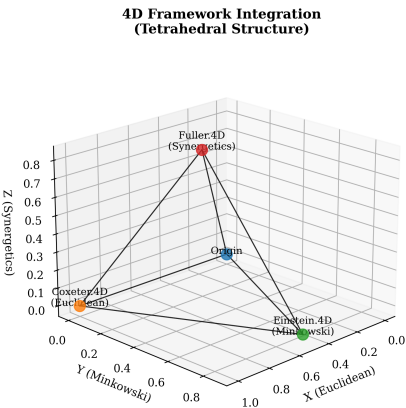
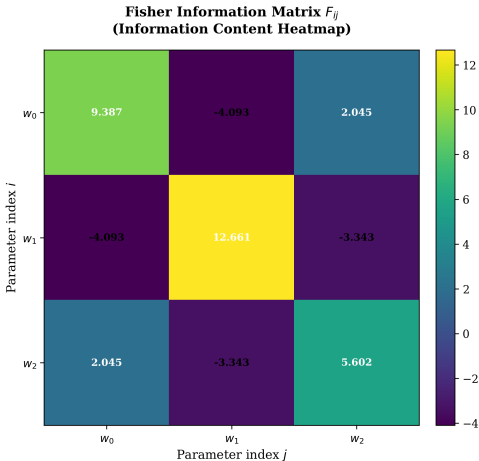
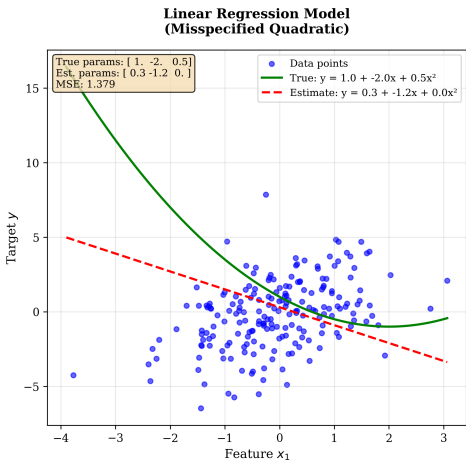
$$L_{\mathcal{n}}$$

$$g_{ij}=F_{ij}\delta_{ij}F^{-1}\nabla L\nabla L$$

$$F^{-1}\nabla L$$

$$\delta_{ij}F_{ij}$$

$$F^{-1}\nabla L\Delta=-\eta F^{-1}\nabla L$$



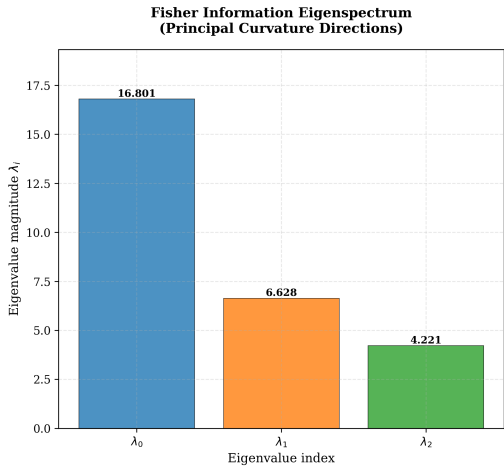
$$y=w_0+w_1x+w_2x^2w_{true}=[1.0,-2.0,0.5]w_{est}=[0.3,-1.2,0.0]F_{ij}F_{ii}w_iF_{ij}w_0,w_1,w_2\delta_{ij}F_{ij}\Delta w=F^{-1}\nabla L$$

$$w$$

$$\eta F\nabla Lg_{ij}=F_{ij}$$

$$F^{-1}\nabla L$$

$$F$$



#### Curvature Analysis & 4D Framework

**Key Metrics:**

**Condition Number:** 3.98  
**Anisotropy Index:** 0.592  
**Total Curvature:** 27.651

**Eigenvalue Interpretation:**

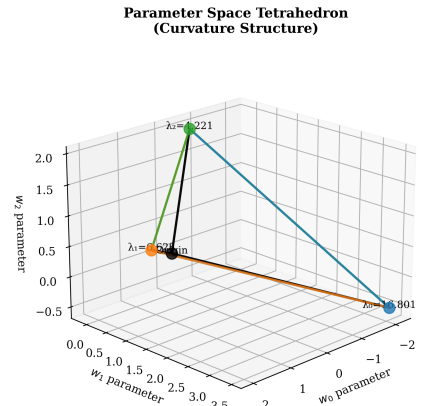
- $\lambda_0 = :.3f$ : Primary direction
- $\lambda_1 = :.3f$ : Secondary direction
- $\lambda_2 = :.3f$ : Tertiary direction

**4D Framework Connection:**

- Coxeter.4D: Euclidean parameter space
- Einstein.4D: Fisher metric geometry
- Fuller.4D: Tetrahedral structure

**Optimization Implications:**

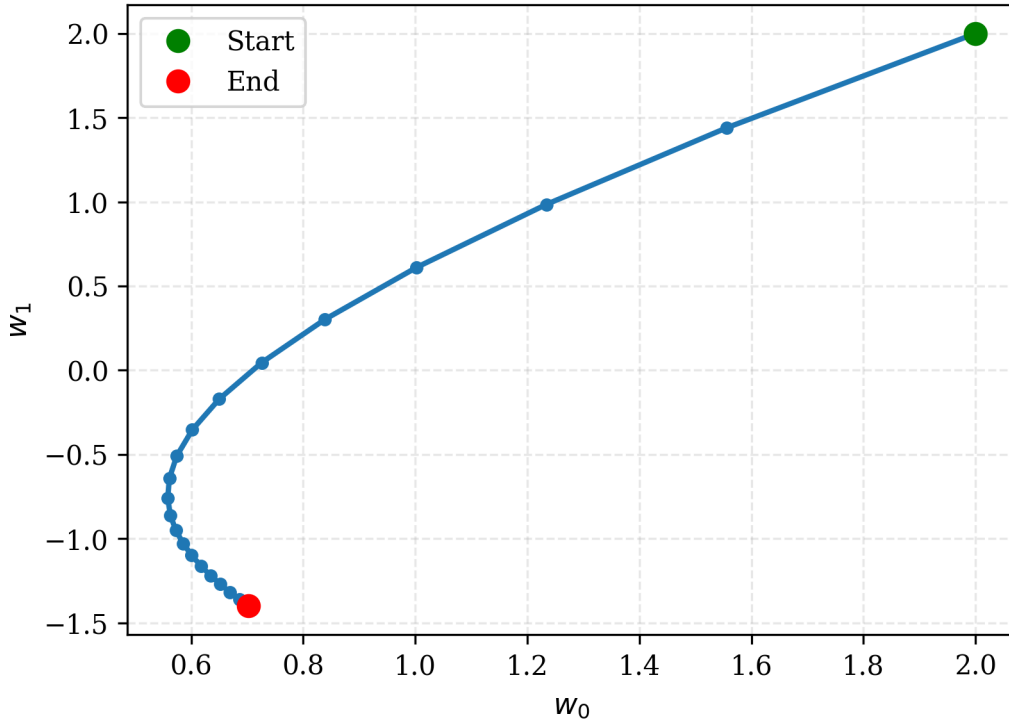
- Natural gradient scales by  $F^{-1}$
- Anisotropic scaling improves convergence



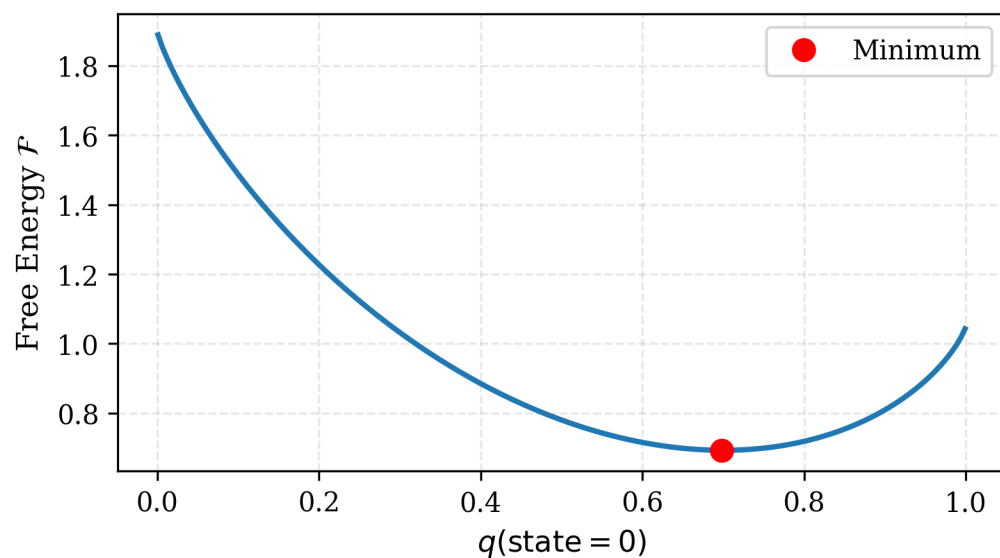
$$\lambda_i F^{-1} F^{-1}$$

#### Natural Gradient Trajectory

(Geodesic Motion on Information Manifold)



### Variational Free Energy Landscape (2-State System)



$$\mathcal{F} = -P(o|s) + [Q(s)||P(s)]q(=0)\mathcal{F}$$

$$\mathcal{F} = - P(o \mid s) + [Q(s) \parallel P(s)]$$







$$V=\frac{1}{6}\parallel P_1-P_0,\;P_2-P_0,\;P_3-P_0\parallel$$

$$P_0,\ldots,P_3^{1/6}$$

$$V_{ivm}=\frac{1}{4}\left|\begin{pmatrix}a_0&a_1&a_2&a_3&1\\b_0&b_1&b_2&b_3&1\\c_0&c_1&c_2&c_3&1\\d_0&d_1&d_2&d_3&1\\1&1&1&1&0\end{pmatrix}\right|$$

$$M(q_0,q_1,q_2,q_3)=\begin{bmatrix}q_{01}&q_{02}&q_{03}&q_{04}&1\\q_{11}&q_{12}&q_{13}&q_{14}&1\\q_{21}&q_{22}&q_{23}&q_{24}&1\\q_{31}&q_{32}&q_{33}&q_{34}&1\\1&1&1&1&0\end{bmatrix},\qquad V_{ivm}=\frac{1}{4}\mid M(q_0,q_1,q_2,q_3)\mid$$

$$V_{xyz}=\frac{1}{6}\left|\begin{pmatrix}x_a&y_a&z_a&1\\x_b&y_b&z_b&1\\x_c&y_c&z_c&1\\x_d&y_d&z_d&1\end{pmatrix}\right|,\qquad V_{ivm}=S3\,V_{xyz},\quad S3=\sqrt{\frac{9}{8}}$$

$$S3=\sqrt{9/8}$$

$$_{288}V^2=\begin{pmatrix}0&1&1&1&1\\1&0&d_{01}^2&d_{02}^2&d_{03}^2\\1&d_{10}^2&0&d_{12}^2&d_{13}^2\\1&d_{20}^2&d_{21}^2&0&d_{23}^2\\1&d_{30}^2&d_{31}^2&d_{32}^2&0\end{pmatrix}$$

$$d_{ij}P_iP_j$$

$$V_{ivm}=S3\cdot V_{xyz}S3=\sqrt{9/8}$$

$$144\,V_{xyz}^2=4a^2b^2c^2-a^2\,(b^2+c^2-f^2)^2-b^2\,(c^2+a^2-e^2)^2-c^2\,(a^2+b^2-d^2)^2+(b^2+c^2-f^2)(c^2+a^2-e^2)(a^2+b^2-d^2)$$

$$\begin{array}{l} a,b,cd,e,f \\ V_{ivm}=S3\cdot V_{xyz}S3=\sqrt{9/8} \end{array}$$

$$V_{ivm}=\frac{1}{4}\left|\begin{pmatrix}a_1-a_0&a_2-a_0&a_3-a_0\\b_1-b_0&b_2-b_0&b_3-b_0\\c_1-c_0&c_2-c_0&c_3-c_0\end{pmatrix}\right|$$

$$P_1-P_0P_2-P_0P_3-P_0$$

$$F_{i,j} = \mathbb{E}\left[\frac{\partial}{\partial \theta_i} p(x;\theta) \frac{\partial}{\partial \theta_j} p(x;\theta)\right]$$

$$F_{i,j} = \frac{1}{N} \sum_{n=1}^N \frac{\partial}{\partial \theta_i} p(x_n;\theta) \frac{\partial}{\partial \theta_j} p(x_n;\theta)$$

$$NN\rightarrow\infty$$

$$\theta \leftarrow \theta - \eta \, F(\theta)^{-1} \, \nabla_{\theta} L(\theta)$$

$$\mathcal{F} = -\, P(o \mid s) + [Q(s) \parallel P(s)]$$

$$G = [Q(s) \parallel P(s)] \; - \; H[Q(s)] \; - \; \mathbb{E}_{\boldsymbol{q}}[\, P(o \mid s) \,] \; - \; P(o)$$

$$\mathbb{E}_{\boldsymbol{q}}[Q(s)] = -H[Q(s)]$$

$$P(o)GG$$

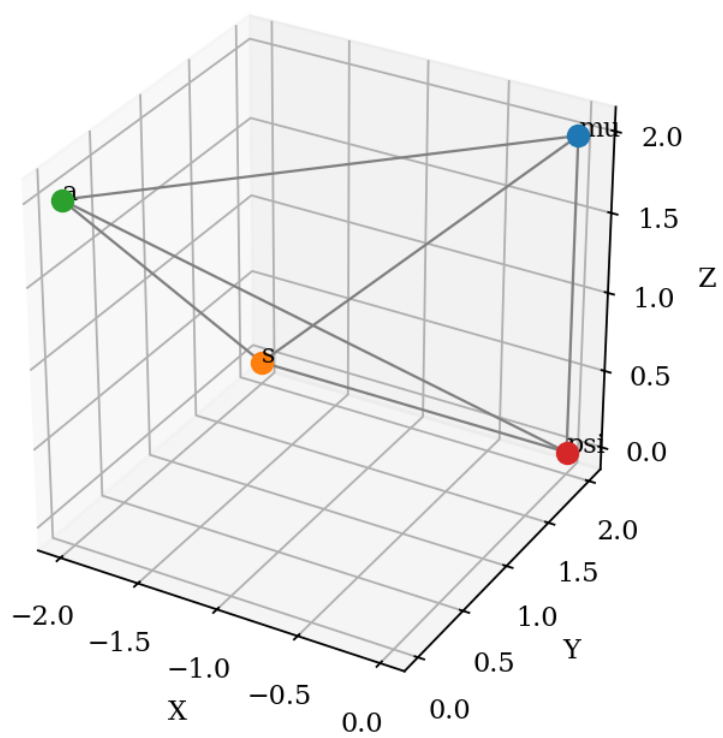
$$q=(a,b,c,d)k=(a,b,c,d)q'=q-(k,k,k,k)$$

$$M\mathbb{R}^3\mathbb{R}^4d(q_1,q_2)=\|M(q_1)-M(q_2)\|_2$$

$$ds^2 = - c^2 \, dt^2 + dx^2 + dy^2 + dz^2$$

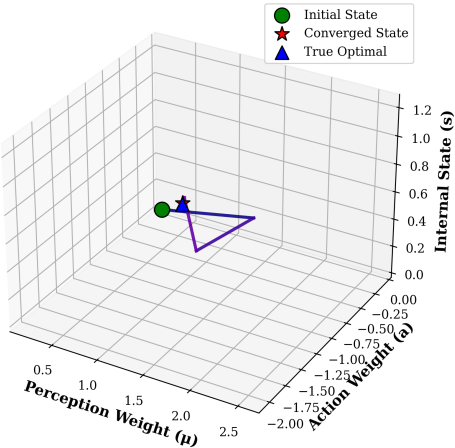
$GG$

Four-fold partition mapped to Quadray tetrahedron

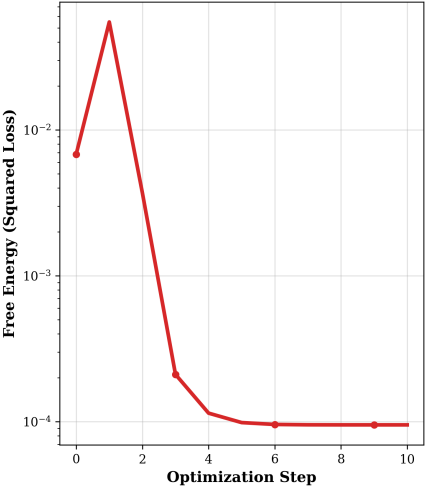


$\mu$   
 $s$   
 $a$   
 $\psi$

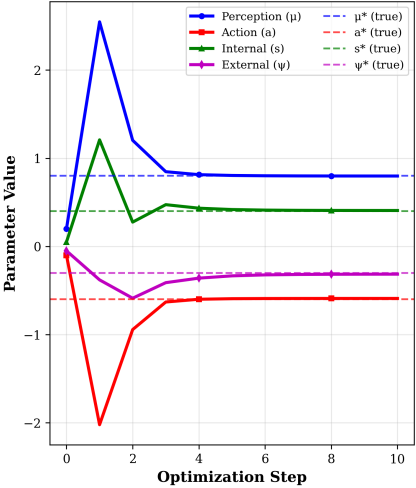
4D Natural Gradient Trajectory  
(Active Inference Dynamics)



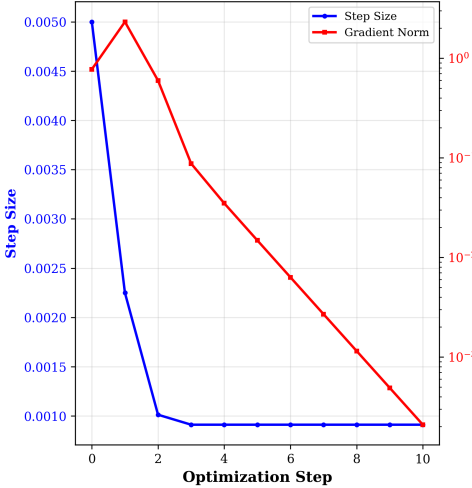
Free Energy Minimization  
(Active Inference Principle)



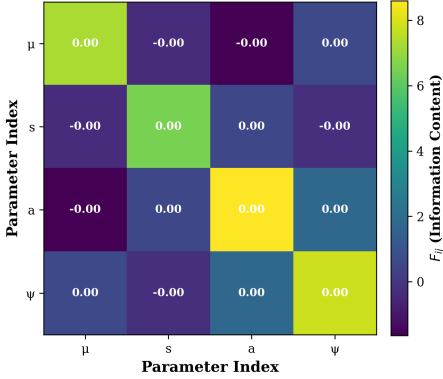
Four-Fold Partition Evolution  
(Fuller.4D Tetrahedral Structure)



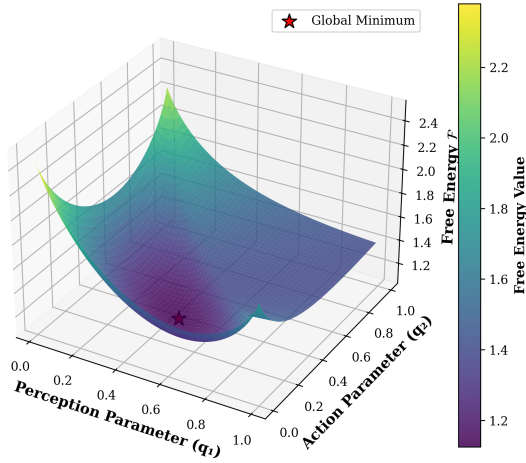
Optimization Diagnostics  
(Step Size & Gradient Evolution)



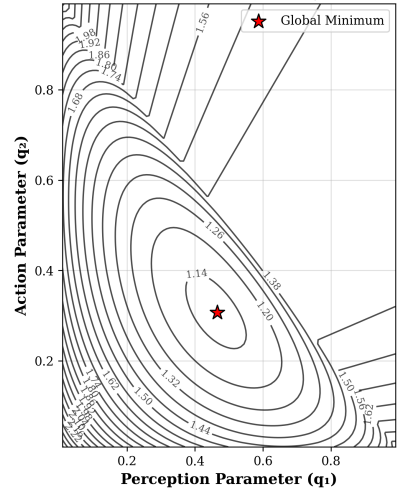
Fisher Information Matrix  
(Information Geometry Metric)



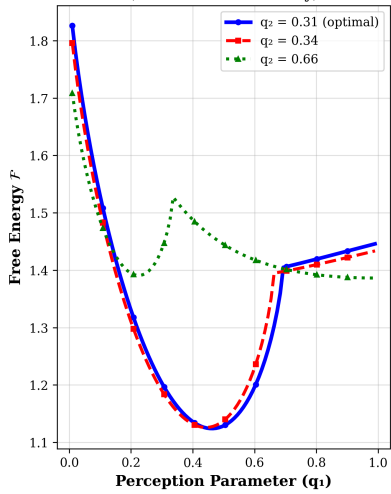
Free Energy Landscape  
(4D Active Inference Framework)



Free Energy Contours  
(Information Geometry)



Free Energy Cross-Sections  
(Parameter Sensitivity)



Active Inference Four-Fold Partition:

$\mu$  (Internal States)  
 $s$  (Sensory)  $a$  (Actions)  
 $\psi$  (External Causes)

4D Framework Integration:

- Fuller.4D: Tetrahedral structure with IVM quantization
- Coxeter.4D: Exact Euclidean geometry for measurements
- Einstein.4D: Information-geometric flows and geodesics

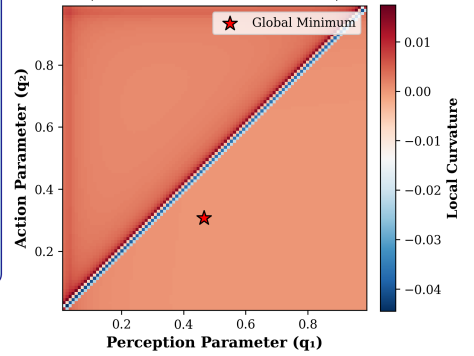
Free Energy Principle:

- Minimization drives perception-action cycles
- Balances prediction error vs. complexity
- Natural gradient follows geodesics on information manifold

Mathematical Structure:

- Variational posterior  $Q(s)$  over latent states
- Prior  $P(s)$  encoding environmental structure
- Likelihood  $P(o|s)$  connecting observations to states

Local Curvature  
(Fisher Information Structure)



$$\frac{\mu s \nabla}{a \psi \nabla_a} \mu^F$$

$$\mathbb{R}^n$$

$$\mathbb{R}^{n_n}$$

$$\begin{array}{l} q=(a,b,c,d)\\ A,B,C,D\\ k\\ q'\\ P_0,\ldots,P_3\\ d_{ij}\\ (\cdot)\\ |\cdot|\\ V_{ivm}\\ V_{xyz}\\ S3 \end{array}$$

$$k=(a,b,c,d)q'=q-(k,k,k,k)$$

$$P_iP_j$$

$$V_{ivm}=1$$

$$S3=\sqrt{9/8}V_{ivm}=S3\,V_{xyz}$$

$$\begin{array}{l} \alpha\\ \gamma\\ \rho\\ \sigma\\ V_{ivm} \end{array}$$

$$\begin{array}{l} \alpha,\gamma,\rho,\sigma\\ V_{ivm} \end{array}$$

$$e$$

$$\mathbb{E}[\cdot]$$

$$F_{ij}$$

$$\mathcal{F}$$

$$[Q\|P]$$

$$\nabla_{\theta}L$$

$$\eta$$

$$\theta$$

$$ds^2$$

$$c$$

$$\mathbb{E}[\partial_{\theta_i} \, p \cdot \partial_{\theta_j} \, p]$$

$$-P(o \mid s) + [Q(s) \parallel P(s)]$$

$$\sum Q\left(Q/P\right)$$

$$F(\theta)^{-1}\nabla_{\theta}L(\theta)$$

$$\theta_i$$

$$-c^2\,dt^2+dx^2+dy^2+dz^2$$

$$M$$

$$\|\cdot\|_2$$

$$R,D$$

$$\mathbb{R}^3$$

$$\sqrt{x_1^2+\cdots+x_n^2}$$

$$RDD=2R$$

$$\alpha,\gamma,\rho,\sigma$$

$$\theta$$

$$\mu$$

$$\psi$$

$$\eta$$